Int J Earth Sci (Geol Rundsch) (2002) 91:838–849
DOI 10.1007/s00531-002-0261-3

ORIGINAL PAPER

M. Schätz · T. Reischmann · J. Tait · V. Bachtadse
H. Bahlburg · U. Martin

The Early Palaeozoic break-up of northern Gondwana, new palaeomagnetic and geochronological data from the Saxothuringian Basin, Germany

Received: 26 March 2001 / Accepted: 26 November 2001 / Published online: 9 April 2002
© Springer-Verlag 2002

Abstract Early Palaeozoic volcanic and sedimentary rocks from the Saxothuringian Basin (Franconian Forest, northern Bavaria) have been subjected to detailed radiometric and palaeomagnetic studies in order to determine the tectonic environment and geographic setting in which they were deposited. Two hand samples were collected from the as yet undated pyroclastic flow deposits for $^{207}Pb/^{206}Pb$ age dating. Radiometric results for these samples, obtained by the single-zircon evaporation technique, are identical within error, and the mean age of all measured grains is 478.2±1.8 Ma (*n*=11). This age is considered to be primary and firmly constrains the eruption of the ignimbrites and formation of the subaqueous pyroclastic flows as having occurred in Early Ordovician (Arenig) times. Palaeomagnetic studies were carried out on these Early Ordovician volcanic rocks, and also on the biostratigraphically dated, Late Ordovician (Ashgillian) Döbra sandstones. The volcanic rocks carry up to three directions of magnetisation. The poorly defined, low and intermediate unblocking temperature directions are thought to represent secondary overprint directions of post-Ordovician age. The high temperature component, however, identified at temperatures of up to 580 °C, is of mixed polarity and passes the fold test with 99% confidence. The overall mean direction after bedding correction is 189°/76°, α_{95}=11.6°, k=44.7 (25 samples, five sites), and is considered to be primary and Early Ordovician in origin. It yields a palaeo-south pole at 24°N and 007°E, which translates into palaeolatitudes of 63°+ 21.7°/–17.3° S for the Saxothuringian Terrane. Samples from the Late Ordovician Döbra sandstone are generally very weakly magnetised. A high temperature D component of magnetisation can be identified in some samples and yields a mean direction of 030°/–58°, α_{95}=18.5°, k=25.7 (15 samples, four sites) after bedding correction. The Arenig palaeomagnetic results indicate high palaeolatitudes, but separation from northern Gondwana. This is in basic agreement with data from elsewhere in the Armorican Terrane Assemblage, all of which suggest high southerly palaeolatitudes in the Early Ordovician. The geochemical signatures of these rocks indicate emplacement in an extensional environment. These new data, therefore, are interpreted as marking the onset of rifting of Saxothuringia from the north African margin of Gondwana, and the start of the relative northward migration of the Saxothuringian Terrane. Although the Late Ordovician palaeomagnetic results presented here are only poorly constrained, they suggest an intermediate palaeolatitude for Saxothuringia in Ashgillian times, in good agreement with Late Ordovician palaeomagnetic data from the Barrandian.

Keywords Saxothuringia · Palaeozoic · Palaeogeography · Palaeomagnetism · Geochronology

M. Schätz · J. Tait (✉) · V. Bachtadse
Institut für Allgemeine und Angewandte Geophysik, Ludwig-Maximilians-Universität, Theresienstrasse 41, 80333 München, Germany
e-mail: jenny@geophysik.uni-muenchen.de
Tel.: +49-89-23944238, Fax: +49-89-23944205

T. Reischmann
Institut für Geowissenschaften, Johannes Gutenberg-Universität, Mainz, Germany

H. Bahlburg
Geologisch-Paläontologisches Institut, Westfälische Wilhelms-Universität Münster

U. Martin
TU Bergakadamie Freiberg, Institut für Geologie, Bernhardt von Cotta Str. 2, 09596 Freiberg, Germany

Introduction

This study examines a subaqueous pyroclastic succession of Early Palaeozoic age from the Randschiefer Series and Vogtendorf Beds, exposed near Kupferberg in Bavaria (Figs. 1 and 2). These rocks, which belong to the Saxothuringian Basin of the Saxothuringian Zone within the Variscan fold belt, are thought to be bimodal in character, but have not yet been described in detail. Likewise, the age of these rocks is also only very poorly constrained, and the geodynamic and palaeogeographic set-

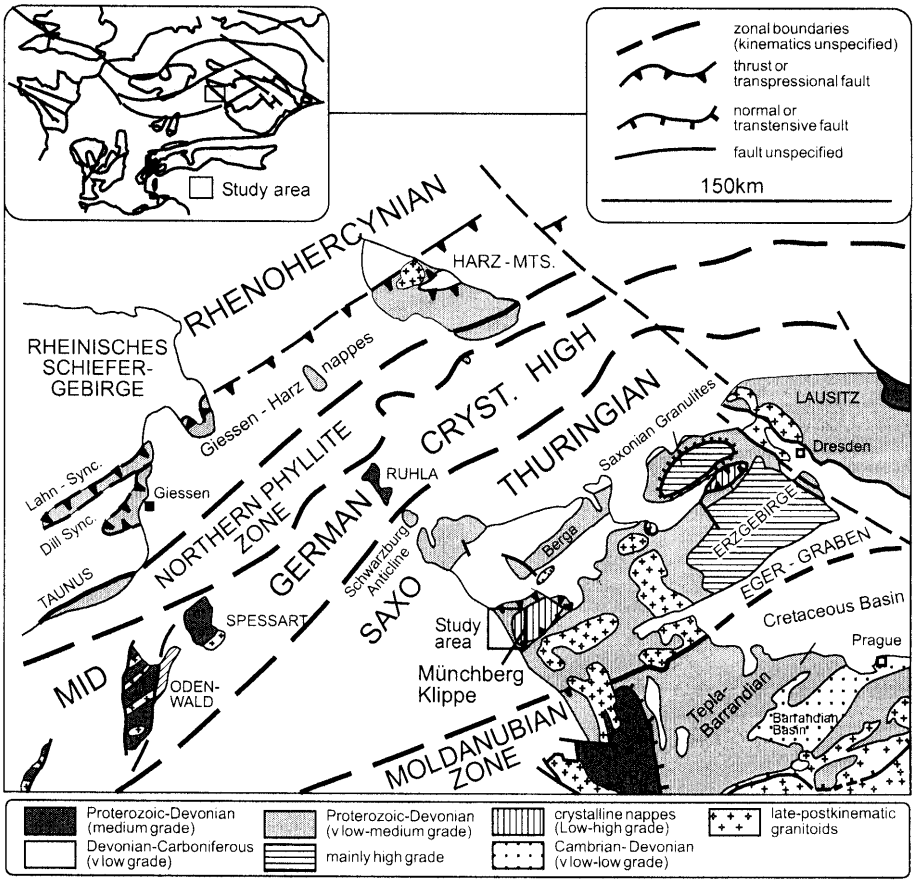


Fig. 1 Sketch map of the main structural and lithological units within the Variscan fold belt of central Europe

ting under which they were deposited can only be loosely inferred. Few palaeogeographically diagnostic faunas have as yet been identified in Saxothuringia, but those which have (trilobites) show certain similarities to Barrandia, Baltica and Gondwana in the Early Ordovician (Sdzuy 1955, 1971).

On geological grounds, it is clear that the Saxothuringian Terrane was part of the Armorican Terrane Assemblage which encompasses Palaeozoic units of the European Variscan fold belt located south of the Rhenohercynian Zone (Avalonia). The presence of Cadomian basement in most units of the Armorican Terrane Assemblage and recent provenance analyses demonstrate they were adjacent to the north African Gondwana margin in the Early Cambrian (Kröner et al. 1995). No palaeomagnetic data have yet been published for the Saxothuringian in Early Palaeozoic times, and for the Early Ordovician few data have been published for the Armorican Terrane Assemblage as a whole. Three reliable Early Ordovician palaeopoles are available for the Armorican Massif (reliable being those which pass the first three of the reliability criteria described by Van der Voo 1990). These data are rather scattered and indicate southerly palaeolatitudes of 55±11° (Cap de Chevre, Cogné et al. 1991), 72±6° (Moulin de Chateaupanne, Perroud et al. 1986), and 62±6° (Pont Rean, Cogné 1988). One additional pole is available from the Prague Basin of the Czech Republic (Barrandia) which indicates palaeolatitudes of 76±14° (Tait et al. 1994). No reliable data are as yet available for Saxothuringia or the Iberian Massif. Early Ordovician sedimentary and volcanic rocks in the Saxothuringian Terrane, however, are suggestive of rifting from northern Gondwana (Linnemann et al. 2000), similar to the scenario indicated by palaeomagnetic evidence from Barrandia.

For Late Ordovician times the picture is even less clear. Only two sets of palaeomagnetic data which pass the above criteria have been published for the Armorican Terrane Assemblage, and these provide contrasting results. Data from Barrandia (Tait et al. 1995) show intermediate palaeolatitudes (approx. 40°S), indicating separation from the northern margin of Gondwana sometime in the early–mid Ordovician. Data from the Vendée province of the Armorican Massif, however, indicate continued peripolar palaeolatitudes (approx. 76°S; Perroud and Van der Voo 1985). No reliable palaeomagnetic data are as yet available for the Late Ordovician of Saxothuringia or Iberia, and thus the affinities and movement of these blocks can only be constrained by faunal and lithological indicators. Late Ordovician sediments of Saxothuringia are characterised by Ashgillian glacio-marine deposits, similar to those found elsewhere in the Armorican Terrane Assemblage and are related to the Saharan glaciation centred in northern Africa. Studies indicate, however, that these diamictites were deposited from floating or seasonal ice (Brenchley et al. 1991) and not from the main ice

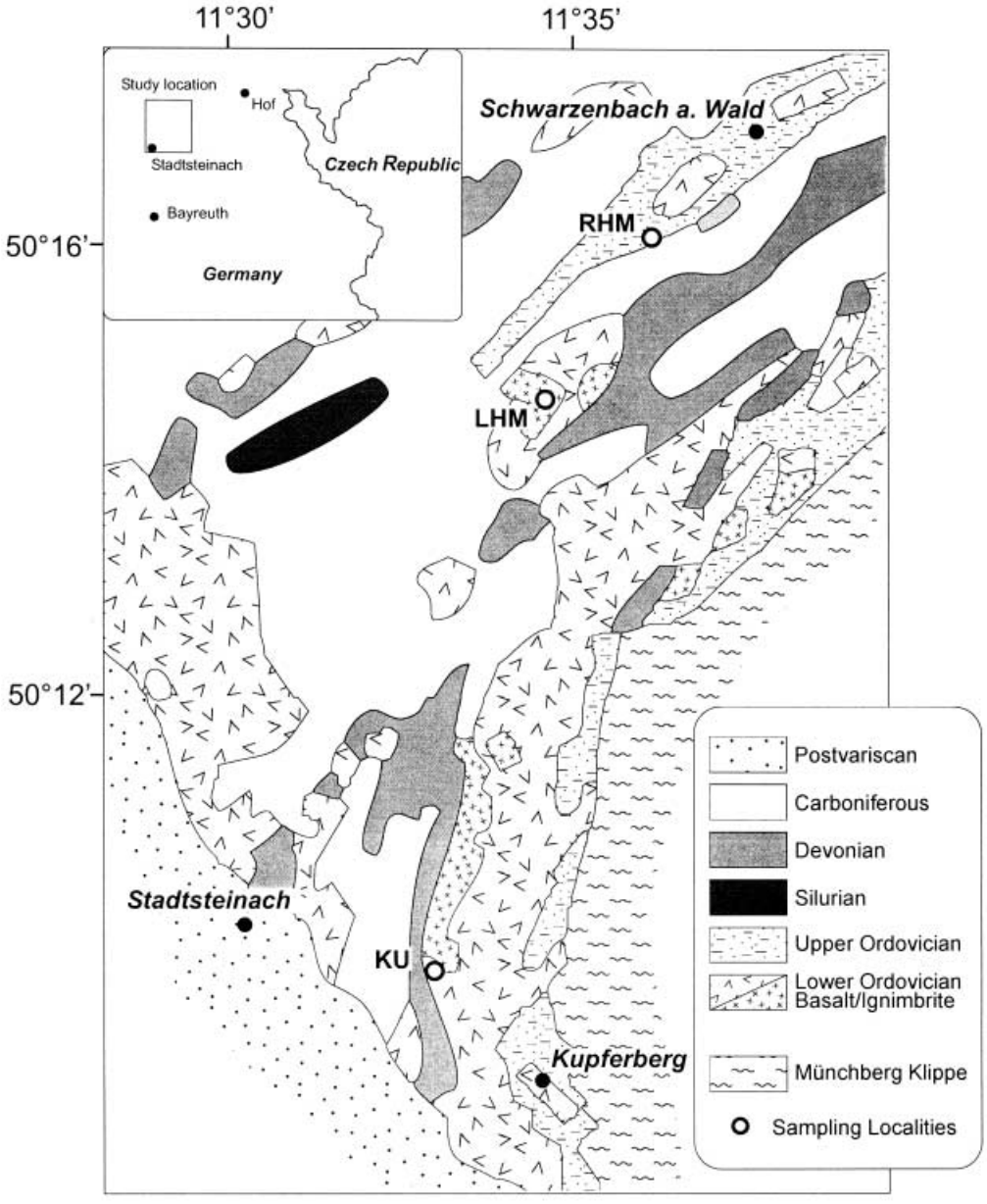


Fig. 2 Simplified geological sketch map of the Franconian Forest, NE Bavaria. *Open circles* represent sampling localities discussed in the text

sheet itself, and that the effects of the Ashgillian cooling may have extended to more intermediate palaeolatitudes (Owen et al. 1991). The presence of glacio-marine deposits, therefore, does not necessarily restrict deposition to a peripolar environment.

Palaeobiogeographic indicators are rather sparse for the Saxothuringian, and the first palaeogeographically indicative fossils (ostracods) for Palaeozoic times are from the Late Silurian Ockerkalk limestones. These show strong similarities to Barrandia (Hansch 1995), thus indicating proximity of Barrandia, Saxothuringia and Armorica at least in Late Silurian times (Kriz and Paris 1982).

To help constrain the palaeogeography of Saxothuringia in Ordovician times, a sequence of volcanic rocks and overlying Late Ordovician sandstones have been subjected to a detailed and multidisciplinary study. Investigations include radiometric dating to determine the formation age of the volcanic rocks, which until now has been constrained only by the underlying Tremadoc and overlying Ashgillian sediments (Sannemann 1955; Sdzuy 1971), and palaeomagnetic studies of volcanic and sedimentary rocks to constrain the palaeogeographic setting.

Geological setting

The Saxothuringian Basin of the Variscan belt in central Europe (Fig. 1) is a north-easterly-trending basin, bounded to the NW by the Mid-German Crystalline Rise (MGCR), and to the south-east by the north-west margin of the Barrandian. This latter boundary is marked by a major fault zone which is largely covered by Permo-

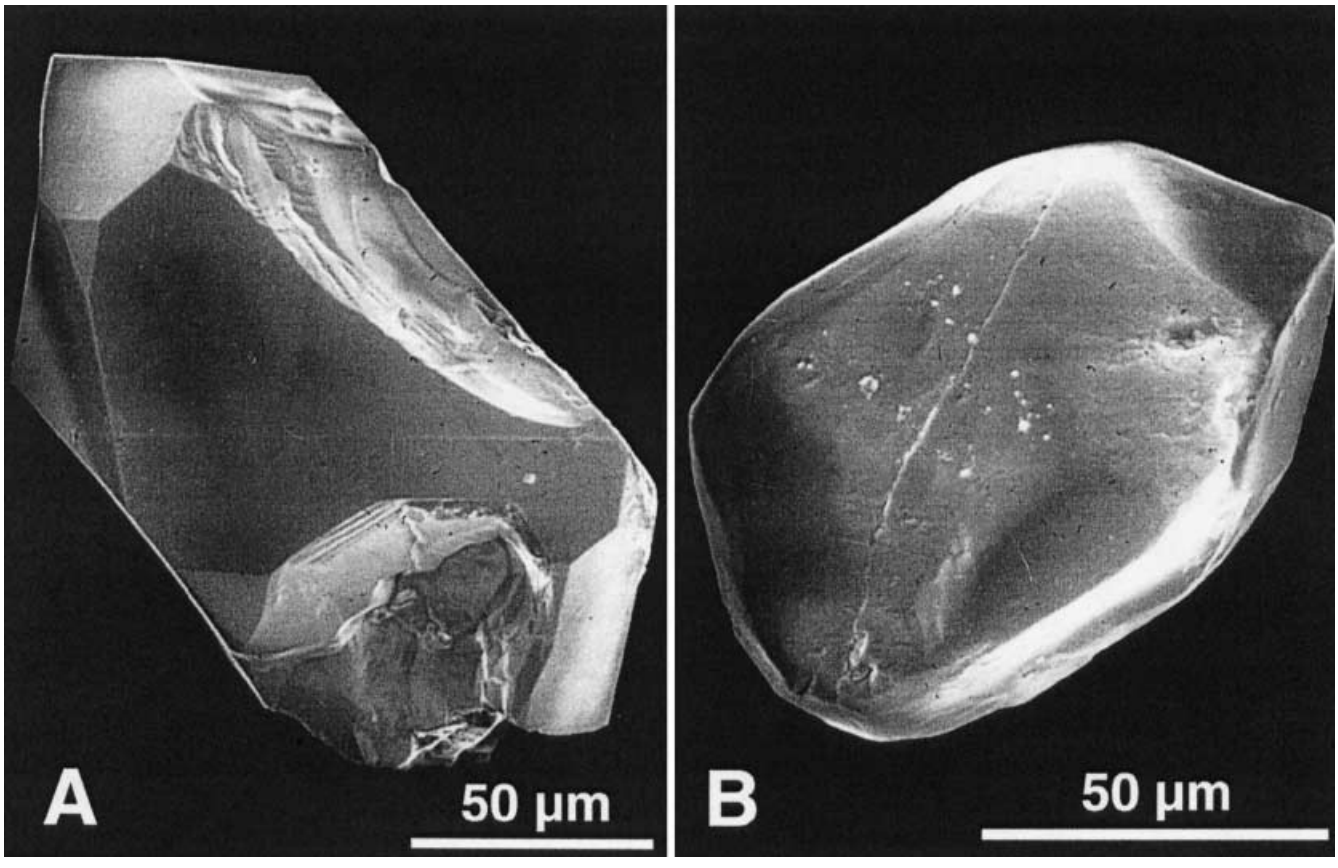


Fig. 3A, B SEM photographs of typical zircons from the Kupferberg ignimbrite. The classification of Pupin and Turco (1975) is indicated. **A** KU-8, grain length is ca. 190 mm, type J4/J5. **B** KU-8, grain length is ca. 110 mm, type S24/S25

Carboniferous deposits (Matte et al. 1990). The Saxothuringian Zone (Fig. 1) is thought to comprise at least two Early Palaeozoic palaeogeographic units, the Saxothuringian Terrane, which is considered part of the Armorican Terrane Assemblage, and the Erzgebirge which shows more affinities to Avalonia. Differentiation between these structural units is based on inherited zircon populations and provenance analyses (Kröner et al. 1995, 2000; Friedl et al. 2000). The MGCR (Fig. 1) is generally thought to have formed the leading edge of the Saxothuringian Terrane during its post-Ordovician evolution (Reischmann et al. 2001; Anthes and Reischmann 2001), and was separated from the Rhenohercynian Zone of the Variscan belt (the southern margin of Avalonia) by the Rheic Ocean. The exact location of this suture zone is not clear, but is thought to be camouflaged within the Northern Phyllite Zone (Fig. 1; Franke and Onken 1995). Episodic closure of the Rheic Ocean is documented by Silurian and Devonian calc-alkaline volcanism in the southern part of the Northern Phyllite Zone and the northern part of the MGCR (Sommermann et al. 1992; Reischmann and Anthes 1996; Anthes and Reischmann 2001).

The Saxothuringian Basin represents a Cambro-Ordovician rift basin which developed on Cadomian crust (Franke et al. 1995; Linnemann et al. 2000). Evidence for MORB-type basalts and ophiolites, which may represent fragments of a Saxothuringian ocean floor, are now preserved as eclogites along the steeply dipping shear zone, or suture, separating this zone from the Moldanubian and Barrandian to the south-east (Matte 1991; Franke et al. 1995). The Saxothuringian Ocean closed in Late Palaeozoic times, and the Saxothuringian Terrane and Barrandian were welded together by the Late Devonian (Franke and Engel 1986).

Two facies associations can be identified in the Saxothuringian Basin and were termed the Thuringian facies and the Bavarian facies by Wurm (1925). As described by Falk et al. (1995), the Bavarian facies consists largely of hemipelagic and pelagic sediments which have been thrust over the (para)autochthonous sequences of the Thuringian facies. The Bavarian facies is exposed predominantly as isolated klippes and nappes of high-grade metamorphic rocks. In the Kupferberg region, however, the Bavarian facies rocks are of very low metamorphic grade. By contrast, the Thuringian facies rocks, exposed in the present-day north-western part of Saxothuringia, are in general of very low metamorphic grade, and comprise neritic to hemipelagic shelf realms, i.e. the shallower water equivalents of the allochthonous units.

The subject of this study are the Early Palaeozoic beds of the Bavarian facies which crop out along the northern margin of the Münchberg Gneiss Massif (Figs. 1 and 2).

Table 1 Single zircon Pb/Pb evaporation data of pyroclastic samples from Kupferberg quarry. *C* Clear, *lr* translucent

	Size (µm)	Colour	Temperature (°C)	Ratio	$^{207}Pb/^{207}Pb$	2σm	Age (Ma)	Error (Ma)
KU7								
A	200	Pink, tr	1,530	75	0.056690	0.000073	479.5	2.8
B	250	Pink, c	1,510	97	0.056646	0.000065	477.8	2.5
Mean KU7				172	0.056668	0.000031	478.7	1.2
KU8								
A	300	Pink, c	1,550	111	0.056629	0.000066	477.1	2.6
B	250	Pink, tr	1,570	55	0.056666	0.000142	478.4	5.5
C	300	Pink, tr	1,530	120	0.056646	0.000078	477.8	3.0
D	300	Pink, tr	1,556	36	0.056655	0.000132	478.1	5.1
E	300	Pink, tr	1,542	180	0.056612	0.000028	476.5	1.1
F	200	Pink, tr	1,550	140	0.056691	0.000051	479.5	2.0
G	300	Pink, c	1,550	90	0.056616	0.000074	476.6	2.9
H	300	Pink, c	1,500	160	0.056693	0.000040	479.6	1.6
I	250	Pink, c	1,520	153	0.056681	0.000061	479.1	2.4
Mean KU8				1,045	0.056654	0.000031	478.1	1.2
Total mean				1,217	0.056657	0.000030	478.2	1.1
Weighted mean (Ludwig 1999)				*n*=11	0.056649	0.000024	477.9	0.9

Geochronology

Two samples of the subaqueous pyroclastic flow deposits (KU7 and KU8), corresponding to the palaeomagnetic sampling sites of the same name, were collected from the Schicker quarry (Fig. 2) and used for radiometric age dating. These pyroclastic horizons occur in the upper part of the outcrop above massive basalts which are, in turn, overlain by pillow basalts. Each hand sample weighs approximately 10 kg, and single-zircon data by Pb/Pb evaporation were obtained following the technique of Kober (1986, 1987). The Pb isotopes of the zircons were analysed at the Max-Planck-Institut für Chemie in Mainz using a Finnigan MAT261 mass spectrometer with a secondary electron multiplier (SEM). Errors for isotope ratios and age calculations of the individual grains are quoted at the 2-sigma level, and ages were calculated using the decay constants of Steiger and Jäger (1977), with common Pb corrections following Stacey and Kramers (1976). Weighted means and errors were calculated according to Ludwig (1999) and are given at a 95% confidence level. Further details concerning sample preparation and analytical procedures are given in Reischmann and Anthes (1996).

The zircons extracted from these samples are all very similar in character and display morphologies typical for a magmatic origin. The grains are bipyramidal and mainly short-prismatic (Fig. 3). The (100) planes clearly dominate the prisms whereas (101) are the prevailing pyramid facets. According to the classification scheme of Pupin and Turco (1975), such zircons belong to S24/S25 and J4/J5 types which are considered to have crystallised at temperatures of 850–900 °C, thus lending further support to the igneous origin of these rocks. The internal structure of the zircons was characterised using cathode-luminescence techniques and shows magmatic zoning as indicated by the variable intensity of the luminescence. Neither core structures nor any other evidence for inheritance could be identified. The crystal morphologies, in combination with their internal structures, indicate that the zircons belong to a single population formed during the magmatic event which produced these pyroclastic rocks.

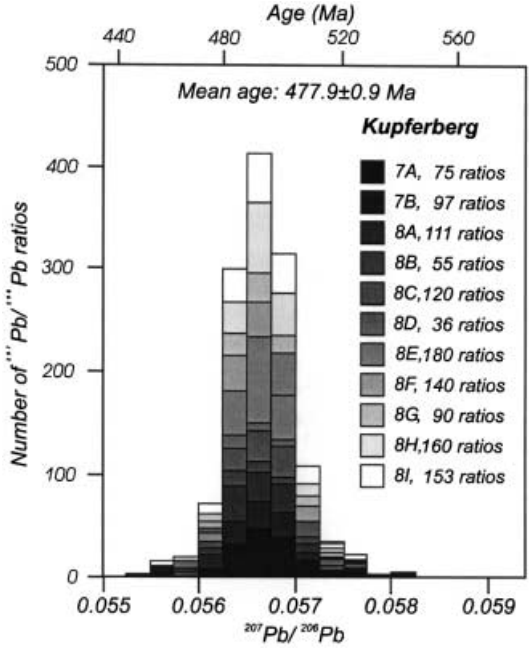


Fig. 4 Histograms showing the results of the $^{207}Pb/^{206}Pb$ single-grain zircon analyses

The zircons analysed for age dating were 200–300 µ in size, mainly clear, and pink or slightly reddish in colour (Table 1). Only grains with no visible intergrowth or inclusions were selected. The $^{207}Pb/^{206}Pb$ ages for all the analysed single-zircon grains are very similar. For the two samples KU7 and KU8 they are 478.6±1.8 Ma (*n*=2) and 477.8±1.1 Ma (*n*=9, Table 1). Being identical within error, these data sets were combined. The weighted mean age of all measured grains from both samples is 477.9±0.9 Ma (*n*=11, Fig. 4), and is interpreted as being

Table 2 Palaeomagnetic site mean directions and overall mean directions obtained for the Early Ordovician volcanic, and Late Ordovician sedimentary rocks

Site	Age	Rock type	Bedding[e]	N/*n*[a]	Dec/Inc (IS)[b]	k[c]	$\alpha_{95°}$[d]	Dec/Inc (TC)[b]	k[c]	$\alpha_{95°}$[d]
KU7	478.2±1.8[f]	Ignimbrites	009/34	11/18	203/50	9.4	19.0	249/80	9.4	19.0
KU8	478.2±1.8[f]	Ignimbrites	354/32	9//4	181/33	92.0	9.6	187/64	92.0	9.6
KU9	478.2±1.8[f]	Ignimbrites	354/32	6/4	163/52	28.7	17.4	112/79	31.7	16.6
KU10	478.2+1.8[f]	Ignimbrites	006/34	5/13	179/36	36.7	20.6	171/70	36.7	20.6
LHM1	Early Ord	Ignimbrites	124/20	11/16	263/64	7.4	26.4	215/73	7.4	26.4
Site mean direction				42/24	190/51	10.7	24.5	189/76	44.7	11.6
RHM1	Ashgill.	Clastics	027/15[g]	5/4	226/40	11.4	28.4	003/–54	11.4	28.4
RHM2	Ashgill.	Clastics	027/15[g]	5/4	206/58	11.0	29.0	029/–73	11.0	29.0
RHM3	Ashgill.	Clastics	027/15[g]	6/4	199/35	16.2	23.6	037/–50	16.2	23.6
RHM4	Ashgill.	Clastics	027/15[g]	5/3	191/34	7.6	48.6	047/–48	7.6	48.6
Site mean direction				21/15	205/42	25.7	18.5	030/–58	25.7	18.5

[a] N is the number of samples measured, and *n* the number of samples used in calculation of site mean
[b] Dec/Inc is the declination/inclination of site means in situ (*IS*) and after tilt correction (*TC*) in degrees
[c] k is the precision parameter
[d] $\alpha_{95°}$ is the half angle of the cone of 95% confidence (Fisher 1953)
[e] Bedding correction is given in terms of direction of dip/dip in degrees
[f] $^{207}Pb/^{206}Pb$ age on single zircons
[g] Beds overturned

the eruption age of the ignimbrite flows. Thus, volcanic activity can be constrained as having taken place in the Arenig, i.e. Early Ordovician. It is important to note that the zircons do not show any indication of secondary Pb loss. This implies that the Variscan deformation and low-grade metamorphism did not affect the U/Pb system of the zircons, although the samples show clear indications of alteration, such as high LOI and some disturbance of the alkali elements. The observed alteration may be associated with very early processes during the submarine eruption and interaction with seawater. HFSEs such as Zr, Hf, Nb, Ta, and Ti are also considered to be undisturbed, since their concentrations follow magmatic variations. This is a further argument that the minerals bearing the age information, namely zircon, and the minerals bearing the magnetic information, namely oxides, were not affected by the presumed seawater alteration and, similarly, have not been affected by any later thermal or thermo-chemical event.

Palaeomagnetism

To constrain the palaeogeographical setting in which these rocks were deposited, samples for palaeomagnetic analysis were collected from two localities (Fig. 2), the Schicker quarry near Kupferberg (sites KU1–KU12), and outcrops close to Schwarzenbach am Wald (LHM1–LHM3). Petrological and geochemical studies show that these sequences have undergone no, or at most only extremely low grades of metamorphism. The deformation in the region as a whole is thought to have peaked in Carboniferous times, associated with the main thermal event identified in this part of the Variscan fold belt at about 340 Ma. Late Ordovician samples were also collected from the Döbra sandstone (RHM1–RHM4) which is exposed near Schwarzenbach am Wald (Fig. 2). These fine-grained sediments are biostratigraphically well constrained (conodonts) as being Ashgillian in age (Sannemann 1955), and were sampled for palaeomagnetic analysis to determine the palaeogeography of Saxothuringia in the Late Ordovician.

Sampling and laboratory procedures

A total of 117 palaeomagnetic cores (one inch diameter) from 19 sites were collected for palaeomagnetic analyses. Twelve sites were in the Early Ordovician volcanic sequence exposed in the quarry near Kupferberg (sites KU1–12; Fig. 2), and three sites in the coeval volcanic rocks which crop out near Stadtsteinach (sites LHM1–3; Fig. 2), thus allowing for a fold test on a regional scale. Four sites (21 samples) were also sampled from the biostratigraphically well-dated Ashgillian Döbra sandstone (sites RHM1–4, Fig. 2). The structural attitude was readily measured at each site. The bedding attitude, lithologies, and palaeomagnetic results for interpretable data sets are listed in Table 2. All samples were taken with a portable, gasoline-powered drill and oriented in the field using a standard magnetic or sun compass whenever possible.

The natural remanence magnetisation (NRM) of the standard, 2.2×2.5 cm samples was measured using a 2G cryogenic magnetometer. Measurements and thermal demagnetisation experiments were all carried out in a magnetically shielded room in the laboratory at the Ludwig-Maximilians-Universität, München. Magnetic susceptibility was measured after each heating step to monitor thermo-chemical alterations. Demagnetisation results were analysed using orthogonal vector plots (Zijderveld 1967), and directions of linear demagnetisation trajectories were calculated using principal component analysis. Statistics of the stable endpoint directions were calculated after Fisher (1953). Isothermal remanent magnetisation (IRM) and thermal demagnetisation of the saturation IRM (SIRM) experiments were also carried out to deter-

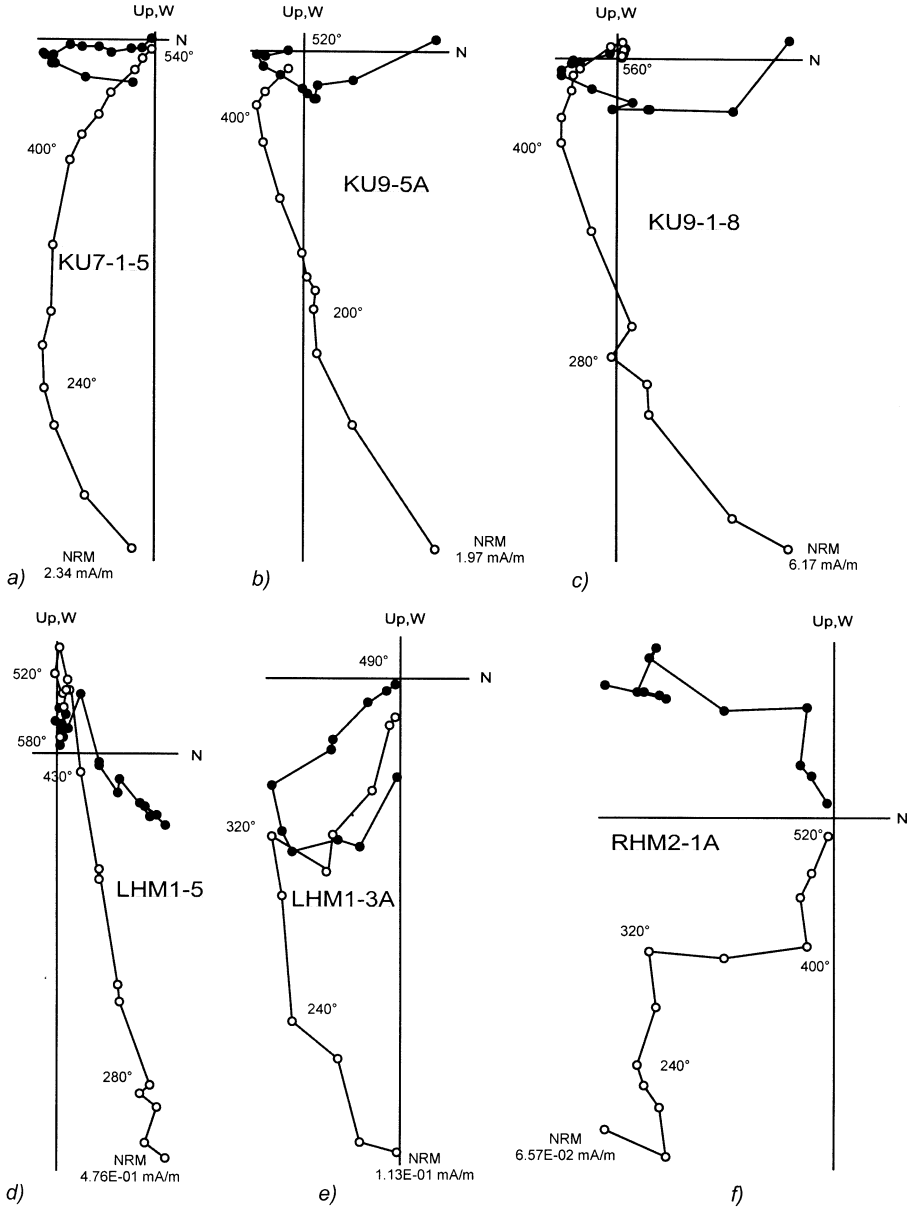


Fig. 5a–f Orthogonal projection of thermal demagnetisation data for **a–e** the Early Ordovician volcanic samples, and **f** the Late Ordovician clastic samples. Directions are plotted in situ, *solid* (*open*) *symbols* represent the horizontal (vertical) component

mine the magnetomineralogy of the samples. In addition, the anisotropy of magnetic susceptibility was measured for a number of volcanic samples. The results show that the rocks overall are essentially isotropic, with maximum anisotropy values for individual samples of 4%, the majority having 1–2% anisotropy.

Palaeomagnetic results

Early Ordovician volcanic rocks

During thermal demagnetisation of samples from the 15 sites (KU1–12, LHM1–3) in the Early Ordovician volcanic rocks, up to two palaeomagnetic directions, termed B and C, can be identified, often after removal of a low unblocking temperature (T_{ub}) A component which generally parallels the local present-day Earth's magnetic field (Fig. 5), and is thought to be of recent origin. Many samples then show removal of an intermediate–high temperature B magnetisation. Although only poorly defined, this component is shallow and southerly directed in situ, and is of single polarity. It often persists until total demagnetisation, and demagnetisation trajectories often follow great circle paths towards a higher temperature direction. No stable endpoint directions for this higher temperature component could be identified, however, and thus further analysis of the great circle paths is not possible. This was the only palaeomagnetic component identified at sites KU1–KU4 which are located in the basal part of the quarry. For these sites, rock magnetic experiments suggest coarse-grained magnetite as the main carrier, and micro- and macroscopic observations suggest relatively high degrees of alteration. Compo-

nent B, therefore, is considered secondary in origin. Due to the uncertainties involved, no further interpretation is given for this component.

Component C can be identified in 25 samples from five sites (KU7–10, LHM1, Table 2) at high unblocking temperatures, generally after removal of direction A and a poorly defined B direction at lower temperatures (Fig. 5a–e). Component C is identified as the stable endpoint direction. It is of dual polarity (Fig. 5), and generally well defined with discrete unblocking temperatures and maximum T_{ub} of 580 °C, suggesting magnetite as the principal magnetic carrier. Direction C is fairly scattered in situ, with southerly to westerly declinations and intermediate to steep inclination values (Fig. 6). There is significant improvement in the statistical distribution of results after bedding correction, however, yielding a steep, southerly mean direction of magnetisation (Fig. 6, Table 2). Isothermal remanent magnetisation acquisition curves of these samples indicate saturation at about 300 to 500 mT (Fig. 7a). A few samples also contain a small amount of a higher coercivity phase (Fig. 7a). Thermal demagnetisation of the SIRM yields high unblocking temperatures of 550–580 °C, thus supporting the interpretation that magnetite is the principal carrier of the magnetisation in these rocks.

Of samples from the remaining six sites sampled in the Early Ordovician sequences (sites KU5–6, KU11–12, LHM2–3), the sample magnetisations yielded no stable direction of magnetisation, were dominated by direction A, or there was no within-site consistency of results. The data obtained from these six sites are not listed in Table 2 and are not discussed further.

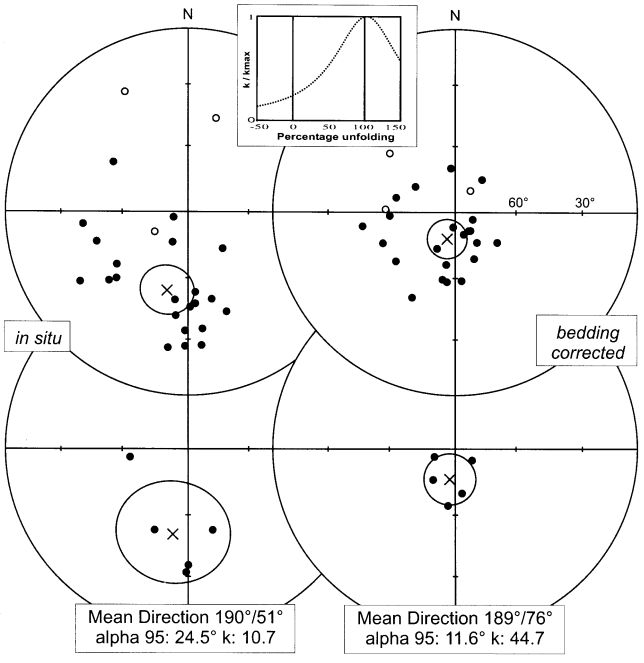


Fig. 6 Equal-area stereographic projections of the Early Ordovician sample (*top*) and site mean (*bottom*) C magnetisation before and after structural correction, and the k/kmax rations versus percentage unfolding. *Solid* (*open*) *circles* represent lower (upper) hemisphere projections; k is the precision parameter (after Fisher 1953), and alpha 95 is the cone of 95% confidence

Upper Ordovician Döbra sandstone

Samples from the Late Ordovician Döbra sandstone (sites RHM1–4) are generally only very weakly magnetised. Nevertheless, it was possible to identify a high-tem-

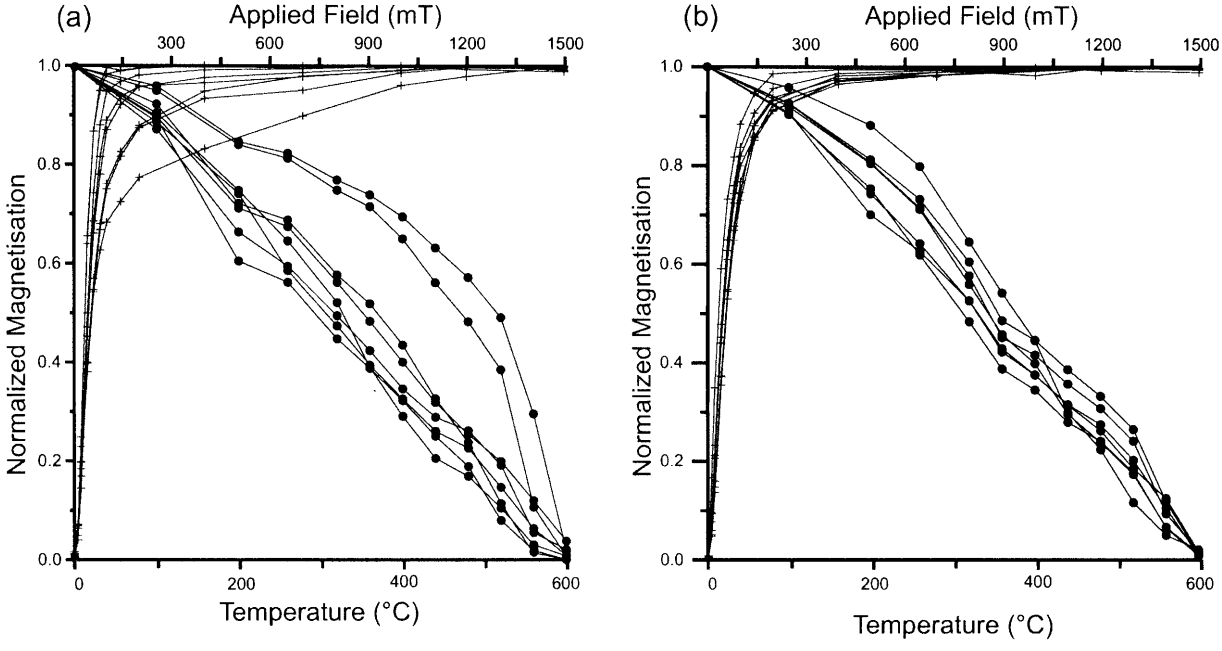


Fig. 7a, b Normalised isothermal remanent (IRM) acquisition curves (*crosses*) and intensity decay plots of the subsequent thermal demagnetisation of the SIRM (*circles*) for **a** the Early Ordovician volcanic samples, and **b** the Late Ordovician clastic sediments

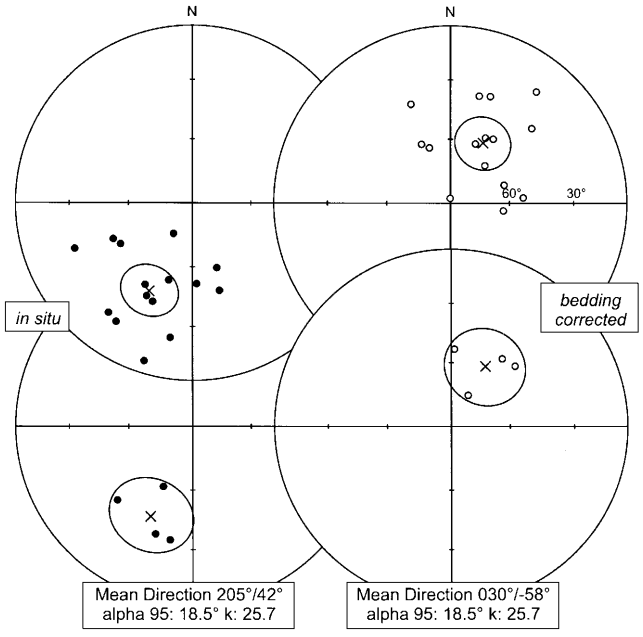


Fig. 8 Equal-area projection of the Late Ordovician sample (*top*) and site mean (*bottom*) D magnetisation before and after structural correction. Notation as for Fig. 5

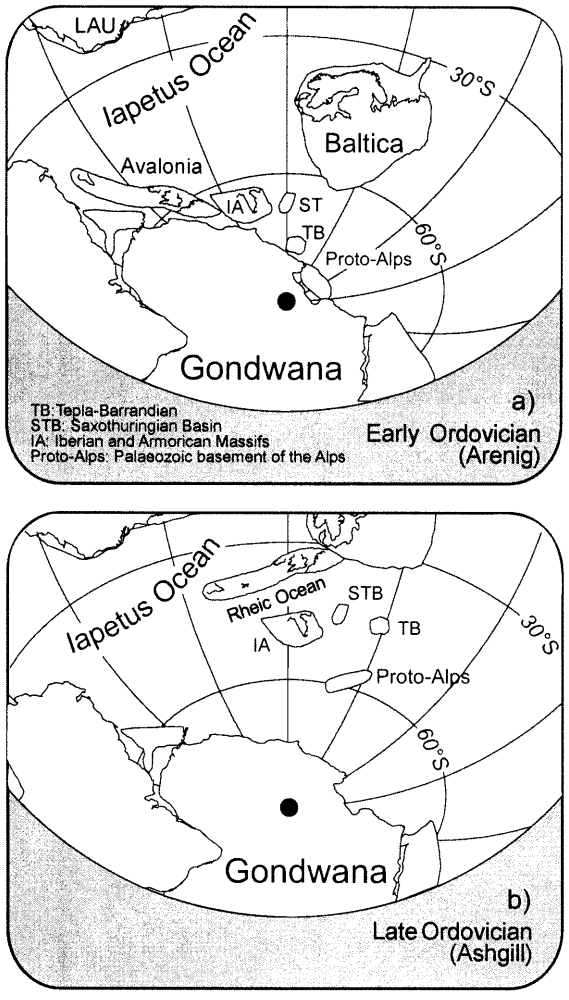


Fig. 9 Palaeogeographic reconstructions for the **a** Early and **b** Late Ordovician based on palaeomagnetic data for the Saxothuringian Terrane (this study), TB (Tepla-Barrandian, Tait et al. 1994, 1995), IA (Iberian and Armorican massifs; see text for details), Baltica (see Smethurst et al. 1998), Gondwana (Bachtadse et al. 1987), and LAU (Laurentia). The Proto-Alps are positioned after Schönlaub (1992)

perature palaeomagnetic direction termed component D (Fig. 5f). Demagnetisation up to 300 °C results in removal of a low-temperature magnetic component, similar to direction A. At intermediate temperatures, some samples show removal of a poorly defined component of magnetisation which is similar in direction to component B (Fig. 5f). Up to the maximum T_{ub}, component D can be identified in a total of 15 samples (four sites), with southerly declinations and intermediate inclination values in situ (Fig. 8). Rock magnetic data (IRM and thermal demagnetisation of SIRM) again demonstrate the predominance of magnetite as the carrier of the magnetisation in these samples (Fig. 7b).

Interpretation and palaeogeographic setting

Direction C, identified in 25 samples (five sites) of Early Ordovician age, is of dual polarity and yields an overall mean direction of 190°/51°, α_{95}=24.5°, k=10.7 in situ, and 189°/76°, α_{95}=11.6°, k=44.7 after bedding correction (Fig. 6, Table 2). This improvement in grouping is statistically significant and passes the fold test of McElhinny (1964) with 99% confidence at 105% unfolding. Component C, therefore, is interpreted as being primary in origin and closely linked to the Early Arenig rock age. The resulting palaeopole position of 24°N 007°E translates into palaeolatitudes of approx. 63°S (Fig. 9) for the Saxothuringian Terrane in Early Ordovician (Arenig) times. This result is similar to those obtained from Barrandia (Tait et al. 1994) and the Armorican Massif (Perroud et al. 1986; Cogné 1988; Cogné et al. 1991), thus demonstrating that, as inferred from geological evidence, Saxothuringia was part of the Armorican Terrane Assemblage (at least in Early Ordovician times) and was at high latitudes, close to the northern margin of Gondwana (Fig. 9).

Component D is identified only in the Late Ordovician age samples, and D yields an overall mean direction of 205°/42°, α_{95}=18.5°, k=25.7 in situ, and 030°/–58°, α_{95}=18.5°, k=25.7 after bedding correction (Fig. 8, Table 2). This component can only be identified in 15 samples and from beds of uniform dip, and thus it is not possible to conduct a fold test to help constrain the relative age of magnetisation. As no palaeomagnetic data from this region have as yet been published, the direction to be expected for Late Ordovician to Late Carboniferous times in this region is not clearly defined. For post-Late Carboniferous times, geological evidence is clear that Saxothuringia was part of Laurasia, and thus expected directions can be estimated from the APW path for Stable

Europe. The in-situ D direction, however, is not similar to any direction which may be expected for post-folding times (i.e. post-Late Carboniferous times). Furthermore, a component interpreted as being a Late Carboniferous overprint direction (component B) is removed at intermediate temperatures, suggesting an older age for component D. Other possible interpretations are either that D is a pre-folding remagnetisation, or it is primary in origin. Given the fact that the bedding-corrected direction is significantly different to that obtained from the Early Ordovician sequences (thus ruling out a widespread pre-folding remagnetisation in the region), and that there is no apparent geological evidence for a local pre-Carboniferous thermal event which may cause a remagnetisation, the D direction is considered to be primary in origin and Late Ordovician in age. Given this interpretation, the inclination values indicate palaeolatitudes of 38±18.5°S for the Saxothuringian Terrane in the Late Ordovician. This is coincident with palaeolatitudes obtained for Ashgillian-age rocks from the Tepla-Barrandian of the Armorican Terrane Assemblage (Tait et al. 1995). However, we are aware that, given the small number of samples in which this component is identified in this study, and the lack of field tests to accurately constrain the relative age of magnetisation, these results for Saxothuringia must remain preliminary until such time as more data become available.

Ordovician palaeogeography of the Armorican Terrane Assemblage and adjacent plates

The tectonic relationships of the various crustal blocks comprising the Armorican Terrane Assemblage (after Tait et al. 1997) remain somewhat arbitrary. From faunal and palaeomagnetic evidence, Barrandia was an autonomous terrane. Relationships between the Armorican and Iberian massifs, and the Saxothuringian Terrane, however, are less well constrained due to the lack of palaeomagnetic and faunal data. In the following discussion and in the palaeogeographic reconstructions presented, Iberia and Armorica are considered to have formed a single crustal block (Ibero-Armorica). It is not clear if Saxothuringia was part of Ibero-Armorica or independent, but that it was separate from Barrandia is fairly clear from geological and palaeomagnetic evidence. The oldest reported palaeogeographically diagnostic fossils for post-Cambrian times are Late Silurian ostracods which indicate close proximity to Barrandia and the Armorican Massif, in agreement with palaeomagnetic data (Kößler et al. 1996). In the following, Saxothuringia is treated as an independent block to highlight the new results presented here, which may or may not be valid for the Armorican and Iberian massifs.

Most elements of Palaeozoic Europe, now juxtaposed between Baltica, Laurentia and Gondwana are characterised by Cadomian crust, and thus were part of the northern Gondwana margin in the earliest Palaeozoic. Age ranges of inherited zircons, however, show a marked difference between Avalonia, indicating a South American provenance, and the Armorican Terrane Assemblage with north African characteristics (Kröner et al. 1995, 2000; Friedl et al. 2000). From palaeomagnetic and faunal evidence, Avalonia (southern Britain) was at 60±5°S in the Arenig (Trench et al. 1992), and adjacent to northern Gondwana (Fig. 9a) until the Llanvirnian when Avalonian faunas show a certain degree of endemism. Tremadocian faunas of the Armorican Terrane Assemblage also remained classically Gondwanan, with the exception of Barrandia whose brachiopod and trilobite genera also show some similarities to those of Baltica (Cocks 2000).

In Arenig times, Barrandia was at 76±14°S (Tait et al. 1994), in agreement with palaeontological data which show continued faunal migration between Barrandia, Armorica and Avalonia (Havlicek et al. 1994). Combining the three poles available for the Armorican Massif places it at 65±20°S (Fig. 9a), i.e. similar to Avalonia and independent of Barrandia. No palaeomagnetic data are as yet available for the Proto-Alps in Ordovician times, the palaeogeographic position of this block being based on faunal and lithological considerations (Schönlaub 1992). The new palaeomagnetic data for Saxothuringia indicate palaeolatitudes of 63° (+21.7°/–17.3°) S in the Arenig, near northern Gondwana. It is interesting to point out that this is concordant with available palaeomagnetic data for Ibero-Armorica. In conjunction with new geochemical data from the volcanic rocks studied (T. Reischmann, unpublished data) the new data presented here are taken as evidence for the onset of rifting of Saxothuringia from Gondwana and the start of its relative northward movement.

For Late Ordovician times, it is clear from faunal and geological information that Avalonia collided with Baltica in the Ashgillian (Cocks et al. 1997), placing it in warmer water latitudes and closing the intervening Tornquist Sea (Fig. 9b). Caradocian faunas of the Armorican Terrane Assemblage show increasing similarity to Avalonia, indicating narrowing of the Rheic Ocean (Fig. 9b) and, by Ashgillian times, Anglo-Baltic faunas can be identified throughout the Armorican Terrane Assemblage. Late Caradocian faunas of Iberia and the Proto-Alps, however, do show some differentiation to those of Barrandia (Havlicek et al. 1994). North African faunas remained of low diversity and cold water in character and distinct to those of Armorica and Barrandia. Palaeomagnetic data for the Armorican Terrane Assemblage are from Barrandia, indicating 40±9.5°S (Tait et al. 1995), the Armorican Massif yielding 76±7°S (Perroud and Van der Voo 1985) and the new data reported here for the Saxothuringian Terrane having palaeolatitudes of approximately 40°S. While a continued high palaeolatitude for the Armorican Massif in the Late Ordovician is in apparent conflict with faunal evidence, it is fairly clear that by Late Ordovician times, Barrandia and Saxothuringia were moving northwards and away from northern Gondwana. Given the faunal record it is tempting to say that the situation for the Armorican Massif

was the same, as shown in Fig. 9b, but until more palaeomagnetic evidence becomes available it's position must remain debatable. This is the case also for the Iberian Massif for which the faunas show a certain degree of endemicity, but for which no reliable palaeomagnetic data have as yet been published.

Acknowledgements The authors gratefully acknowledge financial backing from DFG (MS) and the Volkswagen Foundation (JT). The help of J. Huth, Mainz, during the SEM investigation is also gratefully acknowledged. This is a contribution to the DFG Schwerpunktprogramm "Orogene Prozesse". For further information concerning the geochemical data for these rocks please contact T. Reischmann.

References

Anthes G, Reischmann T (2001) Timing of granitoid magmatism in the eastern mid-German crystalline rise: J Geodyn 31:119–241

Bachtadse V, Van der Voo R, Haelbich IW (1987) Paleozoic paleomagnetism of the western Cape Fold Belt, South Africa, and its bearing on the Paleozoic apparent polar wander path for Gondwana. Earth Planet Sci Lett 84:487–499

Brenchley PJM, Romano TP, Young GC, Storch P (1991) Hirnantian glaciomarine diamictites – evidence for the spread of glaciation and its effect on Upper Ordovician faunas. In: Barnes CR, Williams SH (eds) Advances in Ordovician geology. Geol Surv Can Pap 90-9, pp 325–336

Cocks LRM (2000) The Early Palaeozoic geography of Europe. J Geol Soc Lond 157:1–10

Cocks LRM, McKerrow WS, van Staal CR (1997) The margins of Avalonia. Geol Mag 134:627–636

Cogné J-P (1988) Strain, magnetic fabric, and palaeomagnetism of the deformed red beds of the Pont-Rean formation, Brittany, France. J Geophys Res 93:13673–13687

Cogné JP, Bonhommet N, Kropacek V, Zelinka T, Petrovski E (1991) Paleomagnetism and magnetic fabric of the deformed redbeds of the Cap de la Chèvre formation, Brittany, France. Phys Earth Planet Int 67:374–388

Falk F, Franke W, Kurze M (1995) Stratigraphy. In: Dallmyer RD, Franke W, Weber K (eds) Pre-Permian geology of central and eastern Europe, vol B(1). Springer, Berlin Heidelberg New York, pp 221–234

Fisher RA (1953) Dispersion on a sphere. Proc R Soc A 217:295–305

Franke W, Engel W (1986) Synorogenic sedimentation in the Variscan foldbelt of Europe. Bull Soc Géol Fr ProcUssher Soc 1986(8):25–33

Franke W, Onken O (1995) Zur prädevonischen Geschichte des Rhenohercynischen Beckens. Nova Acta Leopoldina NF 71(291):53–72

Franke W, Dallmeyer D, Weber K (1995) Geodynamic evolution. In: Dallmeyer RD, Franke W, Weber K (eds) Pre-Permian geology of central and eastern Europe. Springer, Berlin Heidelberg New York, pp 579–593

Friedl G, Finger F, McNaughton NJ, Fletcher IR (2000) Deducing the ancestry of terranes: SHRIMP evidence for South America-derived Gondwana fragments in central Europe. Geology 28:1035–1038

Hansch W (1995) The ostracode fauna of the Saxothuringian Ockerkalk sequence (Silurian); new data. In: Int Symp Ostracoda, vol 12. Chapman and Hall, London, 418 pp

Havlicek V, Vanek J, Fatka O (1994) Perunica microcontinent in the Ordovician (its position within the Mediterranean Province, series division, benthic and pelagicassociations). Sbornik Geol Ved 46:23–56

Kober B (1986) Whole grain evaporation for 207Pb/206Pb-age investigations on single zircons using a double-filament thermal ion source: Contrib Mineral Petrol 93:482–490

Kober B (1987) Single-zircon evaporation combined with Pb+emitter-bedding for $^{207}Pb/^{206}Pb$-age investigations using thermal ion mass spectrometry, and implications to zirconology. Contrib Mineral Petrol 96

Kößler P, Tait J, Bachtadse V, Soffel HC, Linneman U (1996) Palaeomagnetic investigations of lower Palaeozoic rocks of the Thueringisches Schiefergebirge. Terra Nostra 96:114–116

Kriz J, Paris F (1982) Ludlovian, Pridolian and Lochkovian in La Meignanne (Massif Armoricain): biostratigraphy and correlations based on Bivalvia and Chitinozoa.: Geobios 15:391–421

Kröner A, Willner AP, Hegner E, Frischbutter A, Hofmann J, Bergner R (1995) Latest Precambrian (Cadomian) zircon ages, Nd isotope systematics and P-T evolution of granitoid orthogneisses of the Erzgebirge, Saxony and Czech Republic. Geol Rundsch 84:437–456

Kröner A, Stipska P, Schulmann K, Jaeckel P (2000) Chronological constraints on the pre-Variscan evolution of the northeastern margin of the Bohemian Massif. In: Franke W, Haak V, Oncken O, Tanner D (eds) Orogenic processes: quantification and modelling on the Variscan Belt. Geol Soc Lond Spec Publ 179:175–198

Linnemann U, Gehmlich M, Tichomirowa M, Buschmann B, Nasdala L, Jonas P, Lützner H, Bombach K (2000) From Cadomian subduction to Early Palaeozoic rifting: the evolution of Saxo-Thuringia at the margin of Gondwana in the light of single zircon geochronology and basin development (central European Variscides). In: Franke W, Haak V, Oncken O, Tanner D (eds) Orogenic processes: quantification and modelling on the Variscan Belt. Geol Soc Lond Spec Publ 179:131–154

Ludwig KR (1999) Isoplot2. Berkeley Geochronology Center Spec Publ 1a:49

McElhinny MW (1964) Statistical significance of the fold test in palaeomagnetism. Geophys J R Astron Soc 8:338–340

Matte P (1991) Accretionary history and crustal evolution of the Variscan belt in Western Europe. Tectonophysics 196:309–337

Matte P, Maluski H, Raijlich P, Franke W (1990) Terrane boundaries in the Bohemian Massif: result of large scale Variscan shearing. Tectonophysics 177:151–170

Owen AW, Harper DAT, Jia-yu R (1991) Hirnantian trilobites and brachiopods in space and time. In: Barnes CR, Williams SH (eds) Ordovician geology. Geol Surv Canada, pp 179–190

Perroud H, Van der Voo R (1985) Palaeomagnetism of the Late Ordovician Thouars Massif, Vendée province, France. J Geophys Res 90:4611–4625

Perroud H, Bonhommet N, Thebault JP (1986) Palaeomagnetism of the Ordovician Moulin de Chateaupanne formation, Vendée, western France. Geophys J R Astron Soc 85:573–582

Pupin JP, Turco G (1975) Typologie de zircon accessoire dans les roches plutonique dioritiques, granitiques et syénitiques. Facteurs essentiels déterminant les variations typologiques. Pétrologie 1:139–156

Reischmann T, Anthes G (1996) Geochronology of the Mid-German Crystalline Rise west of the River Rhine. Geol Rundsch 85:761–774

Reischmann T, Anthes G, Jaeckel P, Altenberger U (2001) Age and origin of the Böllsteiner Odenwald. Mineral Petrol 72(1–3):29–44

Sannemann D (1955) Ordovizium und Oberdevon der bayerischen Fazies des Frankenwaldes nach Conodontenfunden. Neues Jahrb Geol Paläontol Abh 102:1–34

Schönlaub HP (1992) Stratigraphy, biogeography and paleoclimatology of the Alpine Paleozoic and its implications for plate movements.: Jahrb Geol B-A 135:381–418

Sdzuy K (1955) Die fauna der Leimitz-Schiefer (Tremadoc). Abh Senckenberg Naturforsch Ges Frankfurt am Main 492:1–74

Sdzuy K (1971) The Ordovician in Bavaria. BGRM Mem 17:379–389

Smethurst MA, Khramov AN, Pisarevsky S (1998) Palaeomagnetism of the Lower Ordovician Orthoceras Limestone, St. Petersburg, and a revised drift history for Baltica in the early Palaeozoic. Geophys J Int 133(1):44–56

Sommermann A-E, Meisl S, Todt W (1992) Zirkonalter von drei verschiedenen Metvulkaniten aus dem Südtaunus. Geol Jahrb Hesses 120:67–76

Stacey JS, Kramers JD (1976) Approximation of terrestrial lead isotope evolution by a two-stage model. Earth Planet Sci Lett 26:207–221

Steiger RH, Jäger E (1977) Subcommission on geochronology: convention on the use of decay constants in geo- and cosmochronology. Earth Planet Sci Lett 36:359–362

Tait J, Bachtadse V, Soffel H (1994) New palaeomagnetic constraints on the position of central Bohemia during Early Ordovician times. Geophys J Int 116:131–140

Tait J, Bachtadse V, Soffel H (1995) Upper Ordovician palaeogeography of the Bohemian Massif: implications for Armorica. Geophys J Int 122:211–218

Tait JA, Bachtadse V, Franke W, Soffel HC (1997) Geodynamic evolution of the European Variscan fold belt: palaeomagnetic and geological constraints. Geol Rundsch 86:585–598

Trench A, Torsvik TH, Dentith MC, Walderhaug H, Traynor J-J (1992) A high southerly palaeolatitude for Southern Britain in Early Ordivician times: palaeomagnetic data from the Treffgarne Volcanic Formation SW Wales. Geophys J Int 108:89–100

Van der Voo R (1990) The reliability of paleomagnetic data. Tectonophysics 184:1–10

Wurm A (1925) Geologie von Bayern. 1. Nordbayern, Fichtelgebirge, Frankenwald. In: Krenkel (ed) Bodenschätze Deutschlands, vol 2. Borntraeger, Berlin, pp 373

Zijderveld JDA (1967) AC demagnetization of rocks: analysis of results. In: Collinson DW, Creer KM, Runcorn SK (eds) Methods in palaeomagnetism. Elsevier, New York, pp 254–287